Equivalence to Maximization Problem

Because M is symmetric, there exists an orthogonal matrix O and a diagonal matrix D such that $O^{-1}MO = D$, the diagonal components of D are eigenvalues of M that are real numbers, and j-th column vectors of O are eigenvectors corresponding to j-th diagonal element of D. Let $\lambda_1 > \ldots > \lambda_n$ be those eigenvalues. Let also choose D such that its diagonal components are $\lambda_1, \ldots, \lambda_n$ in this order. Suppose we have a vector x such that $\boldsymbol{x}^T\boldsymbol{x} = 1$. Let also $\boldsymbol{y} = (y_1 \cdots y_n)^T = O^{-1}\boldsymbol{x}$. Because O is orthogonal, $O^TO = I$ where I is the elementary matrix, and therefore, we also have $\boldsymbol{y}^T\boldsymbol{y} = \boldsymbol{y}^TO^TO\boldsymbol{y} = \boldsymbol{x}^T\boldsymbol{x} = 1$, that is, the norm of $\boldsymbol{y}$ is also 1. By using these, we can rewrite the given maximization problem as follows.

$$\begin{aligned}
\max_{\boldsymbol{x}^T\boldsymbol{x}=1} \boldsymbol{x}^TM\boldsymbol{x} &= \max_{\boldsymbol{y}^T\boldsymbol{y}=1} \boldsymbol{y}^TO^TMO\boldsymbol{y} \\
&= \max_{\boldsymbol{y}^T\boldsymbol{y}=1} \boldsymbol{y}^TO^{-1}MO\boldsymbol{y} \\
&= \max_{\boldsymbol{y}^T\boldsymbol{y}=1} \boldsymbol{y}^TD\boldsymbol{y} \\
&= \max_{\boldsymbol{y}^T\boldsymbol{y}=1} \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2
\end{aligned}$$

Under the constraint that the norm of $\boldsymbol{y}$ is 1, the formula in the last line takes its maximum value λ_1 when $y_1 = 1$ and $y_i = 0$ $(i \neq 1)$. Because $\boldsymbol{x} = O\boldsymbol{y}$, x that maximizes $\boldsymbol{x}^TM\boldsymbol{x}$ is the first column of O, i.e., the eigenvector corresponding to the largest eigenvalue λ_1.